ISLAMIC UNIVERSITY OF TECHNOLOGY
Department of Computer Science and Engineering (CSE)
Course Outline and Course Plan

Name of the Teacher	S M Sabit Bananee	**Position**	Lecturer
Department	CSE	**Programme**	B.Sc. Eng.
Course Code	CSE 4615	**Course Title**	Wireless Networks
Academic Year	2024-2025	**Semester**	Summer
Contact Hours	2.0	**Credit Hours**	**2.0**
Textbooks and Reference books	1. IEEE 802 Wireless Systems, 1st Edition (Wiley) 2. Wireless Communication Networks and Systems by, 1st Edition 3. Wireless Communications, 2nd Edition 4. Wireless Communications and Networks, 2nd Edition	**Authors of the books**	1. B. H. Walke, S. Mangold and L. Berlemann 2. Beard and Stallings 3. Andrea Goldsmith 4. William Stallings
Prerequisites (If any)	1. CSE 4405 Data and Telecommunications 2. CSE 4511 Computer Networks		
Google Classroom Code	**gjsm3d53**		

Teaching Methods/ Approaches	☑ **Lecture**	☑ **Group discussion**	☐ **Demonstration**	☑ **Problem solving**
	☐ **Project**	☐ **Others: Tutorial classes**		
Teaching aids	☑ **Multi-media**	**OHP**	☑ **Board and Marker**	**Others**

Course Assessment Method								
Attendance (10%)	**Summation of best 3 Quizzes (Total 30%)**				**Extra Evaluation**		**Mid Semester (25%)**	**Semester Final (35%)**
	1st Quiz	**2nd Quiz**	**3rd Quiz**	**4th Quiz**	**Others**			
	Week	**Week**	**Week**	**Week**	**Assignment**	**Homework**	**Week**	**Week**
	3rd Week	**8th Week**	**12th Week**	**14th Week**	**may replace one quiz**	**will be given time to time**	**As per schedule of IUT**	**As per schedule of IUT**
Grading Policy								

Marks out of 100	Letter Grade	Grade Point	Marks out of 100	Letter Grade	Grade Point
80 - 100	A+	4.00	55 - 59	B-	2.75
75 - 79	A	3.75	50 - 54	C+	2.50
70 - 74	A-	3.50	45 - 49	C	2.25
65 - 69	B+	3.25	40 – 44	D	2.00
60 - 64	B	3.00	00 - 39	F	0.00

Course Contents

Introduction to wireless networks: wireless access networks, personal area networks (wireless sensor networks, body area networks, LoWPAN, and Bluetooth), wireless and mobile ad hoc networks, challenged networks (DTNs, VANETs).

Wireless MAC protocols: IEEE 802.11, IEEE 802.11e, IEEE 802.11n, IEEE 802.11s, IEEE 802.15.4, SMAC, B-MAC, IEEE 802.22/20, IEEE 802.16d/e

Wireless routing metrics – ETX, ETT, WCETT, Airtime Metric, routing protocols – AODV, DSR, DSDV, HWMP, sensor network routing, VANET routing, etc.

Wireless Transport protocols: Wireless TCP and its variants, Hop by Hop Congestion Control, Rate based Congestion Control, etc. Quality of Service in Wireless Networks.

Course Objectives

The course objectives of the course Wireless Networks are:

1. To study the fundamentals of wireless communications systems, the wireless network architectures, protocols, and applications.
2. To understand the evolving wireless access technologies and standards.
3. To analyze the performance of various access technologies.

Mapping with CO, PO, and Bloom's Taxonomy

CO No.	Course Outcomes (CO) Statement	levels of Bloom's Taxonomy	Matching with Program Outcome (PO)
CO1	**Identify** and interpret key performance indicators of wireless networks.	C1	PO2
CO2	**Analyze** the operational principles, architectural components, and protocol behaviors of wireless networks using visual and analytical methods.	C4	PO2
CO3	**Apply** the knowledge of wireless networking protocols and architectures to enhance or optimize performance in existing wireless systems.	C3	PO1

	Weekly plan for course content and mapping with CO
Weeks	**Topics**
1	▪ Overview of Wireless Networks, the Internet, and Access Networks ▪ Requirements and Challenges of Various Wireless Access Networks
2-3	▪ Fundamentals of Wireless Communication • Radio Transmission Principles • Radio Spectrum Regulation • Communication Requirements ▪ Basics of Mesh Networking
4-5	▪ IEEE 802.11 Wireless Local Area Networks (WLANs) ▪ MAC Protocols: IEEE 802.11, 802.11e, 802.11n
6-7	▪ Challenged Networks: Delay Tolerant Networks (DTN), Vehicular Ad Hoc Networks (VANET)
	MID TERM EXAMINATION:
8	▪ Operational Overview of V2X (IEEE 802.11p and C-V2X) ▪ Requirements and Challenges of VANET ▪ Wireless Sensor Networks ▪ Bluetooth Technology
9	▪ Wireless and Mobile Ad Hoc Networks (MANET) ▪ Limitations of Traditional MAC Protocols
10	▪ Modern Solutions for MAC Protocols
11-12	▪ High Throughput Networks ▪ IEEE 802.11n, MIMO Technology, OFDM
13	▪ STDMA (Self-Organizing Time Division Multiple Access) ▪ Non-Orthogonal Multiple Access (NOMA)
14	▪ Mobility Management in Wireless Communications ▪ Key Enabler in 5G and Emerging 6G Wireless Communications
15	▪ Quality of Service (QoS) in Wireless Networks ▪ Advanced Topics in Wireless Networking: • Semantic Communication • Applications of Machine Learning and Artificial Intelligence in Wireless Communication
	SEMESTER FINAL EXAMINATION:

	Program Outcomes (POs)
PO No.	**Students graduating from the Bachelor of Science in Computer Science and Engineering (B.Sc. in CSE) program, upon graduation students will have the ability to:**
PO1	**Engineering knowledge:** Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization as specified in K1 to K4 respectively to the solution of complex engineering problems.
PO2	**Problem analysis:** Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences. (K1 to K4)
PO3	**Design/development of solutions**: Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. (K5)

PO4	**Investigation:** Conduct investigations of complex problems using research-based knowledge (K8) and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PO5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an understanding of the limitations. (K6)
PO6	**The engineer and society:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems. (K7)
PO7	**Environment and sustainability:** Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts. (K7)
PO8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice. (K7)
PO9	**Individual work and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
PO10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	**Project management and finance:** Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	**Life-long learning:** Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Mapping of COs and POs [Correlation level 1 for low, 2 for moderate and 3 for high]												
Course Outcomes	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1		3										
CO2		3										
CO3	3											

Student's consulting hour:

Thursday: 2:00PM – 2:30PM

Instructor contact details:

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